

Describing shapes precisely

Geometry helps pupils describe the visible world with precision. In Year 3, they draw 2-D shapes, make and recognise 3-D shapes in different orientations, and use accurate language for sides, faces, edges, vertices and angles.

2-D shapes - flat shapes with sides and vertices



TRIANGLE

3 sides • 3 vertices

angles can have different sizes



RECTANGLE

4 sides • 4 vertices

4 right angles

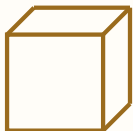


PENTAGON

5 straight sides and 5 vertices

Shapes can be regular or irregular.

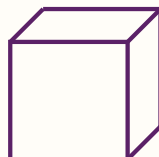
3-D shapes - solid objects with faces, edges and vertices



CUBE

6 square faces

die or building block



CUBOID

6 rectangular faces

book or cardboard box



CYLINDER

2 circular faces

1 curved surface • tin can

Orientation does not change a shape

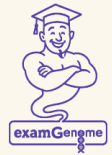
A shape remains the same when it is turned or viewed from another direction. Pupils should identify properties rather than relying only on a familiar position.



Different orientations - still triangles

WHY IT MATTERS

Describing properties develops spatial reasoning for design, construction, art and engineering.



Angles, turns and lines

Angles describe the corners of shapes and the amount of turn. Line relationships describe how lines travel and meet. Both ideas appear constantly in buildings, maps, movement and design.

Angles compared with a right angle

LESS THAN
smaller than 90°

RIGHT ANGLE
exactly 90°

GREATER THAN
larger than 90°

Right angles describe turns

1 **QUARTER-TURN**
 90°

2 **HALF-TURN**
 180°

3 **THREE-QUARTER**
 270°

4 **FULL TURN**
 360°

Example: moving clockwise from 12 to 9 on a clock is a three-quarter turn.

Parallel, perpendicular, horizontal and vertical

PARALLEL

railway tracks

PERPENDICULAR

wall meets floor

HORIZONTAL

level horizon

VERTICAL

upright post

By the end of this topic, pupils should be able to:

- draw 2-D shapes and make 3-D shapes
- recognise 3-D shapes in different orientations
- describe angles as corners and turns
- identify angles greater or less than a right angle
- connect right angles with quarter, half and full turns
- identify horizontal, vertical, parallel and perpendicular lines

FUTURE: Shape properties support area, perimeter, volume, coordinates and transformations.